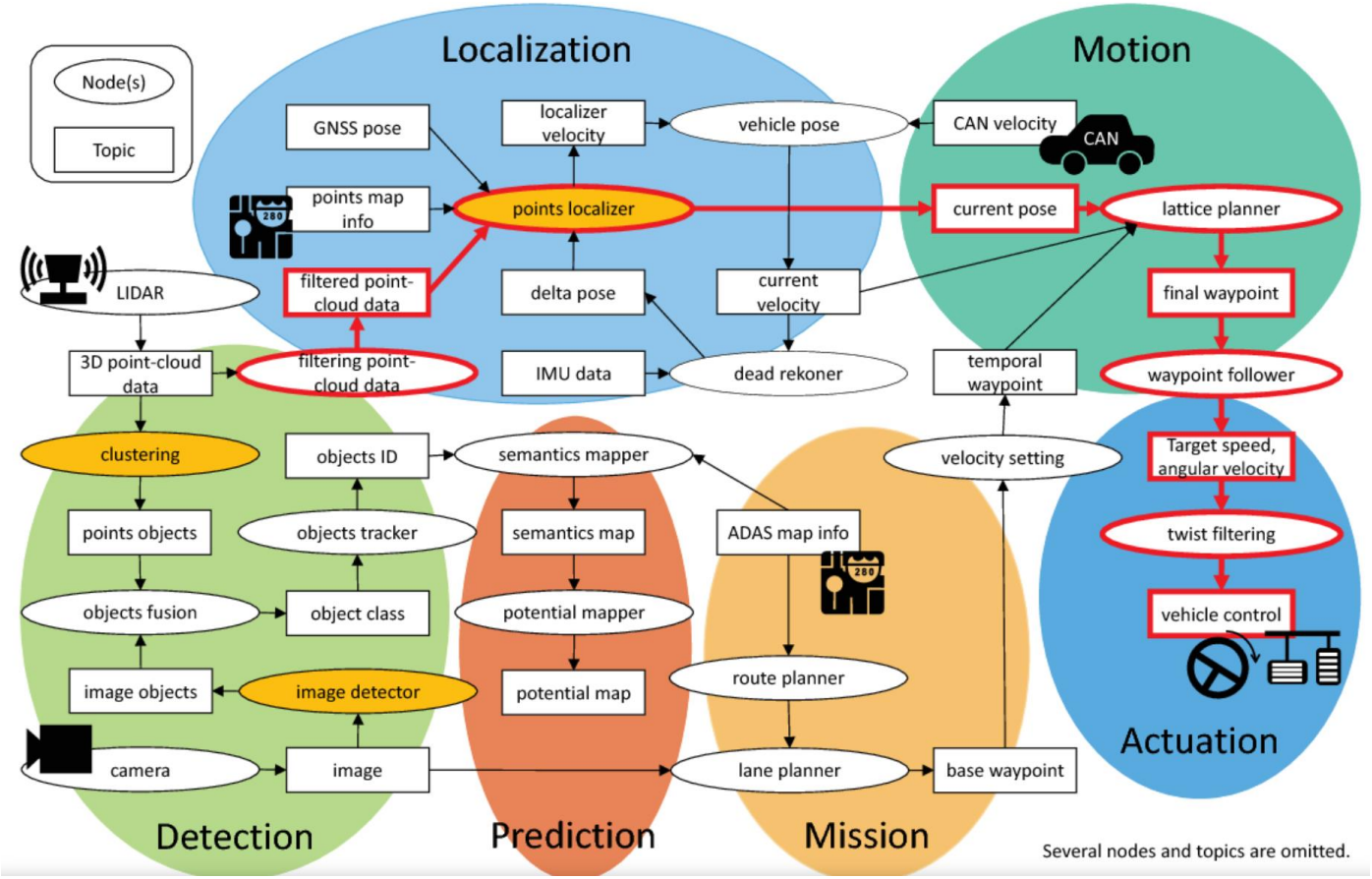
香港城市大學
City University of Hong Kong
Innovating into the Future

Tutorial 2: Apollo Cyber -- Communication

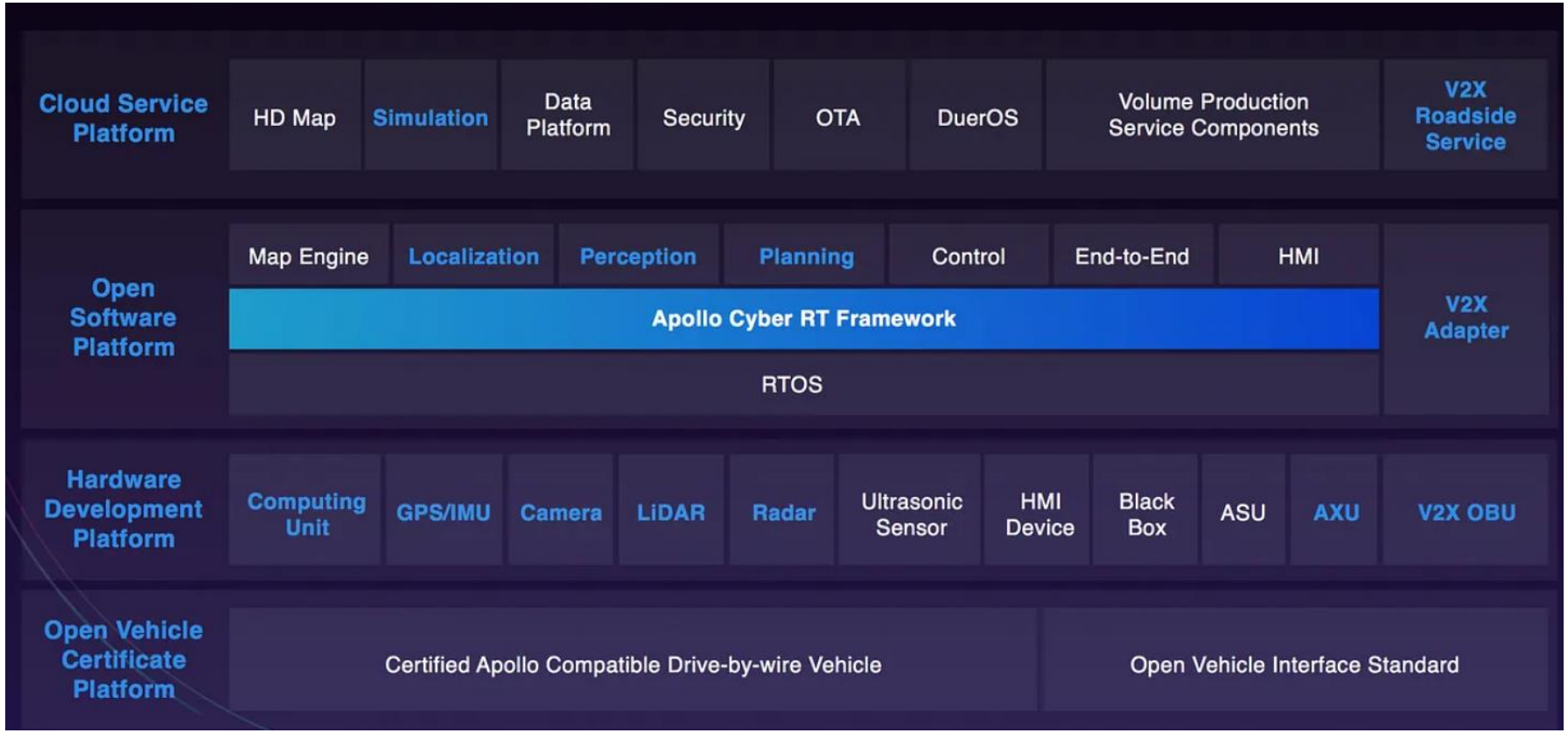
ADS is Complex



Cyber

- A runtime framework for autonomous driving software
- Replaces ROS (Robot Operating System) in Apollo's later versions
- Key characteristics:
  - High performance & low latency
  - Deterministic execution
  - Modular & data-centric
  - Supports heterogeneous hardware
- Built for complex, real-time sensor fusion and decision-making

Cyber


Cloud Service Platform
HD Map
Simulation
Data Platform
Security
OTA
DuerOS
Volume Production Service Components
V2X Roadside Service
Open Software Platform
Map Engine
Localization
Perception
Planning
Control
End-to-End
HMI
Apollo Cyber RT Framework
RTOS
V2X Adapter
Hardware Development Platform
Computing Unit
GPS/IMU
Camera
LiDAR
Radar
Ultrasonic Sensor
HMI Device
Black Box
ASU
AXU
V2X OBU
Open Vehicle Certificate Platform
Certified Apollo Compatible Drive-by-wire Vehicle
Open Vehicle Interface Standard

Communication in Cyber

- Writer/Reader Communication
- Service/Client Communication
- Parameter Services Communication

Writer/Reader Communication

- Publish-subscribe communication model based on shared memory.
- Concepts
  - **Node**: The fundamental unit in the data topology network, capable of creating and managing Writers, Readers, Services, and Clients as needed
  - **Writer**: The publisher in the publish-subscribe model
  - **Reader**: The subscriber in the publish-subscribe model
  - **Channel**: The topic in communication, connecting publishers and subscribers
  - **Message**: The type and data structure used in communication (proto)
- Use Cases
  - One-way communication, where the subscriber does not need to respond to the publisher after receiving a message
  - Scenarios requiring high performance and low latency

Writer/Reader Communication

- **Protobuf**: A cross-language, cross-platform method for serializing data structures; it is a flexible and efficient protocol for serializing data
- **Advantages**:
  - **High performance and efficiency**: After serialization, the byte size is 3 to 10 times smaller than XML, and the serialization speed is 20 to 100 times faster than XML
  - **Ease of use**: Encapsulates operations on structured data into classes, convenient to use
  - **High compatibility**: Both parties in communication use the same data protocol, ensuring that modifications to the data structure by one party do not affect the other
  - **Cross-language support**: Compatible with multiple programming languages, including Java, C++, Python, Go, and more

```
syntax = "proto2";

package apollo.cyber.examples;

message ExamplesConfig {
  optional string car_name = 1 [default = "apollo"];
  required string license = 2;
  repeated string people_name = 3;
}
```

Service/Client Communication

- A C/S communication model encapsulated based on the RTPS protocol
- **Communication Concepts**
  - **Service:** The server in the C/S model.
  - **Client:** The client in the C/S model.
  - **RTPS:** Real-Time Publish-Subscribe protocol
- **Use Cases**
  - Communication between parties not on the same machine.
  - Bidirectional communication, where the client sends request data and the server processes the data and returns response data.

Parameter Services Communication

- Encapsulates the Service/Client communication to achieve a communication method for global configuration parameter sharing
- Concepts:
  - Parameter Server: Stores global parameters
  - Parameter Client: Retrieves and modifies global parameters
- Use Case:
  - Multiple process modules need to share configuration parameters